Mechanical renewal and $D^2(\ln D)^2$ lower bounds for Hilbert numbers

Haibo Lu
Shanghai Institute of Technology, Shanghai 201418, China
luhaibo1985@gmail.com

24 October 2026

Abstract

For every sufficiently large integer D, we construct a real planar polynomial vector field of degree at most D with at least $D^2(\ln D)^2/[8(\ln 2)^2] - O(D^2 \ln D)$ hyperbolic limit cycles. Qin and Sun previously obtained the order $D^2(\ln D)^2$ through two 3-adic filtrations on a Chebyshev center grid. We give a different construction based on auxiliary mechanical folds and planar fourfold renewal, improving the leading coefficient even after polynomial replication extends their subsequence coefficient to all sufficiently large degrees. The auxiliary folds contribute a logarithmic number of birth levels, while planar renewal supplies the second logarithm. Reflections cancel the inherited and opposite-axis periods on each new family, allowing all selected zeros to be realized by one polynomial perturbation. A triangular allocation of monomials closes the total-degree budget, while small even terms fill the intervening degrees. We give an explicit finite lower bound.

1 The problem and the construction

For an integer $D \geq 1$, consider real planar polynomial vector fields

$$\dot{x} = P(x, y), \qquad \dot{y} = Q(x, y), \qquad \max(\deg P, \deg Q) \leq D. \tag{1.1}$$

Let $H(D)$ be the supremum of the cardinalities of finite families of distinct limit cycles carried by a single field (1.1), and let $H_{\mathrm{hyp}}(D)$ be the same supremum counting only hyperbolic cycles. Thus $H(D) \geq H_{\mathrm{hyp}}(D)$. We ask how many cycles can be obtained at a fixed total-degree budget when an old configuration is copied and new cycles are inserted at each stage.

Christopher–Lloyd [1, Section 3] introduced fourfold quadratic replication followed by new cycle nests near axis centers to obtain bounds of order $D^2 \ln D$. Li–Chan–Chung [7, Section 3, Remark 4 and Proposition 5] corrected the axial perturbations in that argument. Itenberg–Shustin [5, Theorem 2.1] obtained an all-degree $D^2 \ln D$ lower bound by a Viro-type gluing construction. Han–Li [3, Lemma 2.6 and Theorem 2.1] developed a two-axis recursion: perturbing the second component by x times an odd polynomial in y creates nests near the nonzero x-axis centers; a smaller odd perturbation in the first component then creates nests near the y-axis centers while retaining the first nests. Their Theorem 1.3 gives the all-degree result

$$\liminf_{D\to\infty} \frac{H(D)}{D^2 \ln D} \geq \frac{1}{2 \ln 2}.$$

Their recurrence is stated for odd-multiplicity cycles, but its births in Lemma 2.6 are hyperbolic. Starting with no selected cycles in the Hamiltonian seeds of their Lemma 3.3, each finite iteration preserves hyperbolicity under the fourfold pullback and the subsequent small perturbations. Thus the same liminf bound holds for H_{hyp}.

Related mechanical ingredients also have direct precedents: Li–Li–Llibre–Zhang [6, p. 445] count two-index monomial periods under a triangular degree constraint for a homogeneous potential; Han–Romanovski [4, Theorems 2.1 and 3.2] combine one-coordinate doubling with new outer cycles in polynomial Liénard systems, already producing one logarithmic accumulation.

The two-source renewal in [8] gives $H_{\mathrm{hyp}}(D) \geq D^2 \log_2 D - O(D^2)$, where the subscript denotes the base. Qin–Sun [9, Theorem 1.1 and Section 6.2] subsequently obtained a lower bound of order $D^2(\ln D)^2$. For every $r \geq 2$ they construct a field of degree at most

$$d_r = 4 \cdot 3^r - 5 \quad \text{with at least} \quad \ell_r = 4r^2 3^{2r-2} - (3^r - 1)^2 \tag{1.2}$$

hyperbolic limit cycles. The coefficient of $d_r^2(\ln d_r)^2$ is $c_{\mathrm{QS}} = 1/[36(\ln 3)^2]$. Their hierarchical Melnikov construction combines two 3-adic filtrations, a local Cartesian-rank argument under sufficiently strong anisotropy, and a simultaneous choice of perturbation coefficients. Within the fourfold copying framework, we construct separable Hamiltonians and perturbing forms directly. We allocate the mechanical monomials according to their degrees after all remaining folds and use the resulting auxiliary history as a source at every planar renewal. This gives the same logarithmic order by a different source mechanism and improves the coefficient. The comparison with the two-source renewal also involves its input class: that step accepts an arbitrary selected hyperbolic configuration, whereas the present construction uses a specially built sequence of separable Hamiltonians. Section 5 compares both the mechanisms and the resulting bounds.

Our main result holds for every sufficiently large integer degree. Throughout the paper, ln denotes the natural logarithm.

Theorem 1.1. *There are absolute constants $C > 0$ and D_0 such that, for every integer $D \geq D_0$, there exists a real planar polynomial vector field of degree at most D having at least*

$$\frac{D^2(\ln D)^2}{8(\ln 2)^2} - CD^2 \ln D$$

distinct hyperbolic limit cycles. In particular,

$$H_{\mathrm{hyp}}(D) \geq \frac{D^2(\ln D)^2}{8(\ln 2)^2} - O(D^2 \ln D). \tag{1.3}$$

Proposition 3.3 gives an explicit finite lower bound $L(n)$ for every integer $n \geq 2$, with field degree at most $2n - 1$. On the dyadic sequence it simplifies to

$$L(2^r) = \frac{4^r r(r+1)}{2} - \frac{8}{3}4^r + 4 \cdot 2^r - \frac{4}{3}, \qquad r \geq 1. \tag{1.4}$$

For $r = 2, 3, 4, 5$, this gives at least $20, 244, 1940, 12756$ cycles at degrees $7, 15, 31, 63$.

The quantitative increment over [8, Proposition 4.1] comes from the source inserted at each stage. That renewal has degree bound $m^+ \leq 2m + 5$ and count $N^+ = 4N + 4(m+2)(m+1)$, whose normalized birth term stays of constant order. Here a mechanical Hamiltonian $z^2/2 + V(w)$ undergoes its own sequence of w folds: each fold doubles the earlier period zeros and creates a

new outer family. We allow both the kinetic-coordinate exponent and the potential-variable exponent to vary. After accounting for the remaining real copies, each auxiliary birth level contributes at quadratic order within the final source budget.

In the planar Hamiltonian $U(x) - V(y)$, the two-coordinate fold then produces four old copies and inserts this source near both axes. On the dyadic sequence, let $n_i = 2^i$ and let K_i denote the number of newly selected period zeros at stage i. Its i auxiliary birth levels give $K_i = in_i^2 - 4(2^i - 1)$. If L_i is the cumulative count, then $L_i = 4L_{i-1} + K_i$, and the normalized count L_i/n_i^2 increases by $i + O(2^{-i})$, rather than by a constant, and summing over i produces the second logarithm. The extra history supplies this factor i; the fold-weighted triangular allocation places the source within the Hamiltonian field degree $2n_i - 1$ and yields the coefficient in (1.3).

There are two obstacles to this counting argument. First, the new sources must coexist with the old ones in a single polynomial form. On every new axis oval a reflection reverses orientation while preserving both the old pullback form and the opposite-axis source. Their periods vanish identically there. The two new amplitudes can therefore be chosen small enough to preserve all old simple zeros without altering the desired new zeros. Second, the two uses of folding must fit the same total-degree budget. A mode born k auxiliary folds earlier has degree in the potential variable $2^k(2q + 1) - 1$; allocating the kinetic-coordinate exponent against this exact cost closes the budget. To cover every integer degree we add small even terms to the potentials. We fix all future auxiliary budgets at the outset, so only finitely many already chosen zeros must survive each addition.

For every integer $D \geq 31$, Qin–Sun's direct all-degree corollary gives a lower bound with leading coefficient $c_{\mathrm{QS}}/9$ and retains the seed hyperbolicity [9, Corollary 1.2 and Sections 6.2, 6.4]. Section 5 uses the replication of Eshkobilov–Kadyrov–Mamayusupov [2] to recover c_{QS} for all sufficiently large degrees, with remainder $O(D^2 \ln D \ln\ln D)$. Thus the ratio of the leading coefficients of the present construction and this all-degree consequence is

$$\frac{1/[8(\ln 2)^2]}{1/[36(\ln 3)^2]} = \frac{9}{2}\left(\frac{\ln 3}{\ln 2}\right)^2 \approx 11.3045.$$

2 Simple period zeros and finite simultaneous choices

We use polynomial one-forms $\omega = A\,dx + B\,dy$, with $\deg\omega = \max(\deg A, \deg B)$. A period zero always refers to a specified connected family of regular compact ovals, parametrized by the Hamiltonian value. Distinct ovals can have the same value. We select pairwise disjoint small closed annuli about the finitely many ovals that will produce cycles.

Lemma 2.1. *Let $f_1, \ldots, f_M$ be linearly independent real analytic functions on a connected open interval I. For every nonempty open subinterval $J \subset I$ there is a real linear combination with at least $M - 1$ distinct simple zeros in J.*

Proof. For analytic functions linear independence implies that the Wronskian is not identically zero. One way to see this is to perform elimination on their convergent Taylor series at an interior point: independent analytic germs admit a basis with strictly increasing orders of vanishing, whose Wronskian has a nonzero leading term. The identity theorem transfers independence to every open subinterval. Choose a point h_0 there where the Wronskian is nonzero. The map from coefficients to the Taylor coefficients through order $M - 1$ at h_0 is invertible.

Fix distinct real numbers $t_1, \dots, t_{M-1}$ and prescribe those Taylor coefficients to be the coefficients of $\prod_{j=1}^{M-1}(h - h_0 - \varepsilon t_j)$. The resulting coefficient vector stays bounded as $\varepsilon \to 0$. For the corresponding function F_ε,

$$\varepsilon^{-(M-1)} F_\varepsilon(h_0 + \varepsilon t) = \prod_{j=1}^{M-1} (t - t_j) + O(\varepsilon)$$

in C^1 on each fixed compact t-interval. Each simple root persists, and its energy lies in the prescribed subinterval for sufficiently small $\varepsilon > 0$. For $M = 1$ the assertion is empty. □

Lemma 2.2. *For a finite family of regular compact Hamiltonian annuli and fixed smooth one-forms, selected simple period zeros persist under a sufficiently small smooth change of the Hamiltonian and forms. A single smallness threshold can be used for the whole finite family.*

Proof. On each compact annulus choose a transverse coordinate and smoothly parametrize nearby level curves; regularity gives this parametrization by the implicit function theorem. Differentiating the integral in the energy parameter shows C^1 convergence of the period functions when the Hamiltonian and forms converge in sufficiently high finite differentiability norms on a neighborhood of the annulus. For each simple zero choose a closed energy interval J on which $|M'| \geq a_J > 0$ and the two endpoint values have opposite signs and absolute value at least $b_J > 0$. Taking the minimum of these finitely many positive margins preserves both properties under a common C^1 perturbation. The intermediate value theorem and monotonicity give a simple zero in each selected interval. The annuli remain disjoint under a sufficiently small perturbation. □

The common threshold in this lemma is taken only after the degree and all the finite forms under consideration have been fixed. No threshold uniform in the degree is used.

The following first-order criterion is standard; see, for example, [6, p. 443]. We record its finite simultaneous form.

Lemma 2.3. *Suppose that a polynomial Hamiltonian H and a polynomial one-form $\omega = A\,dx + B\,dy$ have N selected simple period zeros in pairwise disjoint compact regular annuli. For all sufficiently small nonzero real μ, the single polynomial field*

$$X_\mu = (H_y - \mu B,\ -H_x + \mu A) \tag{2.1}$$

has at least N distinct hyperbolic limit cycles and degree at most $\max(\deg H - 1, \deg \omega)$.

Proof. In the energy coordinate on a local transversal the return displacement is $\mu M(h) + O(\mu^2)$ in C^1, where $M(h) = \oint_{H=h} \omega$ is oriented by the Hamiltonian flow. This expansion follows by smooth dependence of the flow and return time and differentiation at $\mu = 0$. Apply Lemma 2.2 to the displacement divided by μ, using the minimum of the finite margins. At each resulting fixed point the derivative of the return map is $1 + \mu M'(h) + O(\mu^2)$ and differs from 1 for sufficiently small nonzero μ. It is positive and close to 1, so the cycle is hyperbolic. The degree follows directly from (2.1). □

3 A mechanical construction with a finite auxiliary history

The construction uses an indefinite separable Hamiltonian. Two new families of compact ovals occur near the coordinate axes after each quadratic fold. The periods of each source form vanish on the opposite family by reflection. The additional logarithm comes from an auxiliary mechanical history: at a given global stage, all earlier auxiliary birth levels contribute at the same leading order after their real copies are counted. For arbitrary degrees, the potentials are padded by small positive even monomials. We fix the complete finite list of future auxiliary budgets before making these perturbations.

The period calculation uses the monomial powers of [6, p. 445] and the outer-period asymptotics of [4, Lemma 2.3]. We give the required version for arbitrary even potentials, since each fold changes the potential.

Lemma 3.1 (Mechanical outer periods)**.** *Let W be an even real polynomial of degree $2s$ with positive leading coefficient. On the outer period annulus of $F(z,w) = z^2/2 + W(w)$, the periods*

$$I_{p,q}(h) = \oint_{F=h} z^{2p+1} w^{2q}\, dw, \qquad p \geq 0, \quad 0 \leq q < s,$$

are linearly independent for every finite collection of distinct pairs (p,q). A collection of M such forms therefore has a linear combination with $M-1$ simple period zeros in any prescribed nonempty open interval of the outer annulus.

Proof. Write the leading term of W as cw^{2s}, $c > 0$. Scaling w by $(h/c)^{1/(2s)}$ in the integral between the two outer turning points gives

$$I_{p,q}(h) \sim C_{p,q} h^{p+1/2+(2q+1)/(2s)}, \qquad C_{p,q} \neq 0.$$

For a consistent oval orientation the constants have a fixed sign: apart from that sign, the scaled integral is a positive multiple of $\int_{-1}^{1} (1-u^{2s})^{p+1/2} u^{2q}\, du$. The exponents are distinct, since their equality would imply $p - p' = -(q-q')/s$ with $|q-q'| < s$. The term with greatest exponent excludes every nontrivial finite linear relation. The periods are analytic on the connected outer energy interval, so their independence holds on each open subinterval. Lemma 2.1 gives the stated simple zeros. Each zero can be assigned a compact regular annular neighborhood and a positive C^1 persistence margin. □

Reflection cancellation of inherited periods is already used in the classical fourfold construction [1, equation (3.3)]. The following observation records its persistence under the small Hamiltonian changes needed here.

Lemma 3.2 (Symmetry on continued ovals)**.** *Let F be a smooth planar Hamiltonian satisfying $F \circ R = F$ on a neighborhood of a compact regular annular family, where R is a coordinate reflection preserving each oval of the family. If $G \circ R = G$ and G is sufficiently close to F in C^3 on a slightly larger annulus, the nearby regular ovals of G are also preserved by R. If $R^*\alpha = \alpha$, the period of α on this entire nearby family is identically zero.*

Proof. Use the flow of $\nabla F/|\nabla F|^2$ to write an invariant annular collar as (θ, h) with $F = h$. The flow is equivariant under R. For G close to F, the equation $G(\theta, u) = h$ has a unique solution $u = u(\theta, h)$ in this collar, by the implicit function theorem. Uniqueness and $G \circ R = G$ make

its closed normal graph invariant under R. It is a regular closed level component, hence an actual periodic oval. A reflection reverses its oval orientation, and consequently

$$\oint \alpha = \oint R^*\alpha = -\oint \alpha = 0.$$

This holds for every energy in an interval, so it includes the energy derivative of the period. The same collar construction gives the C^1 continuity of periods used below, as in Lemma 2.2. □

Proposition 3.3 (Mechanical renewal)**.** *Fix an integer $n \geq 2$ and put*

$$r = \lfloor \log_2 n \rfloor, \qquad n_i = \left\lfloor \frac{n}{2^{r-i}} \right\rfloor \quad (0 \leq i \leq r),$$
$$\eta_i = n_i - 2n_{i-1} \in \{0, 1\} \quad (1 \leq i \leq r).$$

For $1 \leq i \leq r$ and $0 \leq k < i$, set $e_{i,k} = n_{i-1} \bmod 2^k$ and

$$K_i = i(n_i^2 - \eta_i) - 4\sum_{k=0}^{i-1}(e_{i,k}^2 + \eta_i e_{i,k}) - 4(2^i - 1). \tag{3.1}$$

There are even polynomials U_r, V_r of degree $2n$ with positive leading coefficients and a polynomial one-form ω_r of degree at most $2n-1$ such that its period functions on compact regular annuli of $H_r(x, y) = U_r(x) - V_r(y)$ have at least

$$L(n) = \sum_{i=1}^{r} 4^{r-i} K_i \tag{3.2}$$

simple zeros with pairwise disjoint annular neighborhoods. In particular, there exists a real polynomial vector field of degree at most $2n-1$ having at least $L(n)$ hyperbolic limit cycles. The bound satisfies

$$L(n) \geq \frac{1}{2}n^2r^2 + \frac{1}{6}n^2r - \frac{22}{3}n^2. \tag{3.3}$$

Proof. We construct the Hamiltonian and form by a finite induction, retaining the auxiliary forms needed at later stages. The estimate (3.3) is proved in Section 4.

Auxiliary budgets. For each future global stage $t \in \{1, \ldots, r\}$, we build an auxiliary form through levels $j = 0, \ldots, t-1$ for each of the potentials U_j and V_j. Denote either such form by

$$\Omega_j^{(t)} = \sum_{p \geq 0} z^{2p+1} R_p(w)\, dw.$$

The final pullback into a stage-t axis source has degree at most $2(p + \deg R_p + 1)$, so its endpoint budget is $p + \deg R_p \leq n_t - 2$. Reversing the remaining folds gives the budget at level j:

$$p + 2^{t-1-j}(\deg R_p + 1) - 1 \leq n_t - 2 \quad \text{for each nonzero } R_p. \tag{3.4}$$

A fold $w \mapsto w^2 - b$ replaces R_p by $2wR_p(w^2 - b)$, with $\deg(2wR_p(w^2-b)) + 1 = 2(\deg R_p + 1)$ for $R_p \neq 0$. Thus it preserves (3.4) at the next level. A birth at level ℓ uses exactly the monomials

$$z^{2p+1}w^{2q}\, dw, \qquad 0 \leq q < n_\ell, \qquad 0 \leq p \leq n_t - 1 - 2^{t-1-\ell}(2q+1). \tag{3.5}$$

They satisfy the budget. Their periods are independent by Lemma 3.1.

Set $U_0(w) = V_0(w) = w^2$ and $\omega_0 = 0$. For every label t, choose the level-zero auxiliary form in the span of the monomials (3.5), with one fewer selected simple period zero than the dimension of that span, and with compact regular annular neighborhoods around those zeros. For a one-dimensional space this requirement is empty. The complete list of labels is fixed at this point.

Suppose that stage $i-1$ has been completed. Its data are the actual potentials, one global form and all its selected zeros, and the auxiliary forms $\Omega_{i-1}^{(t)}$ for $t \geq i$, separately for U and V. Every selected zero has a compact regular annular neighborhood and a positive C^1 margin. The list is finite.

Real copies and the two mechanical sources. Write $U = U_{i-1}$, $V = V_{i-1}$, and first consider

$$\widehat{H}_i(x,y) = U(x^2-a) - V(y^2-b), \qquad \Phi(x,y) = (x^2-a, y^2-b).$$

For sufficiently large positive a, b, the four real inverse branches of Φ cover all selected old global annuli. They give four diffeomorphic copies of each old simple period zero for $\Phi^*\omega_{i-1}$, up to orientation. The maps $(z,w) \mapsto (z, w^2-a)$ and $(z,w) \mapsto (z, w^2-b)$ likewise have two diffeomorphic inverse branches on the selected U- and V-auxiliary annuli, respectively. Pulling back the auxiliary forms preserves their period functions up to orientation, so every selected auxiliary zero has two simple copies.

The current vertical source uses the already fixed V-auxiliary form $\Omega_{i-1,V}^{(i)}$. Put

$$c_a = \sqrt{2U'(a)}, \qquad z = c_a x, \qquad w = y^2 - b.$$

On either real y branch the Hamiltonian deficit is exactly

$$U(a) - \widehat{H}_i = K_a(z) + V(w), \qquad K_a(z) = U(a) - U\left(-a + \frac{z^2}{2U'(a)}\right). \tag{3.6}$$

Since U is even with positive leading coefficient, $U'(a) > 0$ for large a. Taylor expansion gives $K_a(z) = z^2/2 + o(1)$ in every fixed C^j norm on compact z intervals. More precisely, if $d = \deg U$, the coefficient of z^{2k} for $k \geq 2$ is $O(a^{-(k-1)d})$. Every regular mechanical oval is individually fixed by $z \mapsto -z$: over its w interval it consists of the two graphs $z = \pm\sqrt{2(h - V(w))}$ joined at their turning points. This holds also for a potential well centered at a nonzero value of w. The C^j convergence of K_a to $z^2/2$ gives continuations of the auxiliary ovals and their simple period zeros to (3.6), by Lemma 2.2. Lemma 3.2 preserves the reflection symmetry of these continued ovals. Pull back the fixed auxiliary form to the polynomial source

$$\alpha_V = \sum_p (c_a x)^{2p+1} R_p(y^2-b)\, 2y\, dy. \tag{3.7}$$

Its periods on the two branches are C^1 close, up to orientation, to the selected auxiliary periods. This produces two copies of every auxiliary simple zero.

Interchanging U, x, a with V, y, b gives a horizontal source

$$\alpha_U = \sum_p (c_b y)^{2p+1} \widetilde{R}_p(x^2-a)\, 2x\, dx, \qquad c_b = \sqrt{2V'(b)}, \tag{3.8}$$

with two copies of the corresponding U-auxiliary zeros. The vertical approximation uses only large a and fixed previous V-annuli; b only has to cover their w ranges. The horizontal approximation uses large b and fixed previous U-annuli, while a covers their w ranges. Thus these are compatible lower bounds on a, b. In particular, no approximation is required near the new outer barriers created by the present folds.

All these annuli can be made disjoint. Old global copies have $|x| \sim \sqrt{a}$, $|y| \sim \sqrt{b}$; the vertical sources have $x = O(1/c_a)$ and $|y| \sim \sqrt{b}$; the horizontal sources have $y = O(1/c_b)$ and $|x| \sim \sqrt{a}$. Within a source, disjointness comes from the selected auxiliary annular neighborhoods. The ovals are genuine closed level components, obtained in regular annular collars; properness of the indefinite Hamiltonian is not needed.

Positive even padding. Fix such a, b, and define

$$U_i(x) = U(x^2 - a) + \epsilon x^{2n_i}, \qquad V_i(y) = V(y^2 - b) + \delta y^{2n_i}, \qquad H_i = U_i - V_i. \tag{3.9}$$

If $\eta_i = 0$, set $\epsilon = \delta = 0$. If $\eta_i = 1$, choose both coefficients strictly positive and sufficiently small. The resulting potentials have exact degree $2n_i$, are even, and have positive leading coefficients.

The finite conditions on this choice are the persistence of the four old global copies, the two auxiliary copies for every label $t > i$, and the current vertical and horizontal source zeros for the fixed forms (3.7) and (3.8). All are conditions on already fixed compact sets and forms. In vertical source coordinates the new error is exactly

$$-\epsilon(z/c_a)^{2n_i} + \delta(w + b)^{n_i}.$$

For fixed a, b, c_a its C^3 norm on each selected compact rectangle tends to zero with (ϵ, δ). In horizontal coordinates the error is $\epsilon(w + a)^{n_i} - \delta(z/c_b)^{2n_i}$ and has the same property. The auxiliary and old global errors also tend to zero on their fixed collars. Lemma 2.2 therefore imposes only finitely many compatible upper bounds on the positive pair (ϵ, δ).

The continued vertical ovals are invariant under $R_x(x, y) = (-x, y)$, and the horizontal ovals under $R_y(x, y) = (x, -y)$, by Lemma 3.2. Both reflections leave H_i invariant. Functoriality of pullback gives

$$R_x^*\Phi^*\omega_{i-1} = R_y^*\Phi^*\omega_{i-1} = \Phi^*\omega_{i-1}, \qquad R_x^*\alpha_U = \alpha_U, \quad R_y^*\alpha_V = \alpha_V.$$

Consequently, on the entire vertical annuli the periods of $\Phi^*\omega_{i-1}$ and α_U vanish exactly. On the entire horizontal annuli the periods of $\Phi^*\omega_{i-1}$ and α_V vanish exactly. Padding changes the old copied periods by a small amount, but does not change any of these fresh-source identities.

Auxiliary births and the global form. With the padded potentials now fixed, complete the auxiliary forms for each remaining label $t > i$. Start with the one-coordinate pullback of $\Omega_{i-1}^{(t)}$, whose selected simple zeros have survived. On a new outer annulus of $z^2/2 + U_i(w)$ or $z^2/2 + V_i(w)$, this pulled form has identically zero period: it is invariant under $w \mapsto -w$, which reverses the oval orientation. In the span of (3.5), choose a fresh birth form with one fewer selected simple period zero than the dimension of that span. Add it to the pulled form with one sufficiently small nonzero scalar. The sum retains every selected inherited zero and supplies the chosen new simple zeros on an outer interval. For either padded potential $W = U_i$ or V_i, choose the new outer interval above all real critical values of W and all values of $z^2/2 + W(w)$

on the previously selected compact annular neighborhoods. Such an interval exists: these neighborhoods are finite in number and compact, and W has positive leading coefficient. Each mechanical level there is a unique outer oval, so the new annular neighborhoods are disjoint from those previously selected. The degree $2n_i$ and positive leading coefficient give the required rank, however small the padding coefficient. The forms for different future stages are chosen independently; these choices change neither H_i nor the current global form.

Finally define the single global form

$$\omega_i = \Phi^*\omega_{i-1} + \lambda_V \alpha_V + \lambda_U \alpha_U. \tag{3.10}$$

Choose two nonzero scalars sufficiently small to retain all four old global copies. On a fresh vertical annulus its period is exactly $\lambda_V \oint \alpha_V$, and on a fresh horizontal annulus it is exactly $\lambda_U \oint \alpha_U$. Thus all new zeros coexist for this same coefficient vector. Each newly chosen scalar is fixed before the next stage: the resulting positive, possibly very small, simple-zero margins are added to the induction data. No uniform margin over the future degree or over all polynomial budgets is required. This completes the finite induction.

Degrees and number of zeros. For an auxiliary monomial with w-coefficient degree d, the vertical or horizontal source degree is at most

$$(2p+1) + 2d + 1 = 2(p+d+1).$$

The final auxiliary budget for label i is $p+d \le n_i - 2$, so each source has degree at most $2n_i - 2$. If $\deg \omega_{i-1} \le 2n_{i-1} - 1$, then

$$\deg \Phi^*\omega_{i-1} \le 2(2n_{i-1} - 1) + 1 = 4n_{i-1} - 1 \le 2n_i - 1.$$

Also $\deg dH_i = 2n_i - 1$. These estimates prove every required degree bound for (3.10) and the final field.

At stage i, put $s = n_{i-1}$, $N = n_i = 2s + \eta_i$. For an auxiliary birth with k remaining folds let $A = 2^k$, $d = \lfloor s/A \rfloor$, and $e = s - Ad = e_{i,k}$. Its actual birth half-degree is d. The allowed modes in (3.5) number

$$M_k = \sum_{q=0}^{d-1} \{N - A(2q+1)\} = d(N - Ad).$$

All summands are positive: at the largest q, their value is $2e + \eta_i + A \ge 1$. In particular $M_k \ge 1$. The A real copies contribute $A(M_k - 1)$ simple zeros. The exact identity

$$AM_k = (s-e)(s+\eta_i+e) = s(s+\eta_i) - e^2 - \eta_i e$$

shows that four source copies give $4\sum_{k=0}^{i-1} A(M_k - 1) = K_i$, as in (3.1). This also proves $K_i \ge 0$ and integrality. The selected-zero counts obey $L_0 = 0$, $L_i = 4L_{i-1} + K_i$, yielding (3.2).

Write $\omega_r = P\,dx + Q\,dy$. Lemma 2.3, applied to the finite disjoint configuration, supplies one sufficiently small nonzero μ for which

$$X_\mu = ((H_r)_y - \mu Q,\ -(H_r)_x + \mu P)$$

has a hyperbolic limit cycle near each selected simple period zero. This is one real polynomial vector field, of degree at most $2n - 1$. □

4 The finite estimate and all degrees

The construction gives $L_i = 4L_{i-1} + K_i$, $L_0 = 0$, with K_i as in (3.1); hence (3.2). We now control the losses uniformly over all binary prefixes. Put $s = n_{i-1}$ and $A_* = 2^{i-1}$. Since $A_* \le s < 2A_*$,

$$\sum_{k=0}^{i-1} e_{i,k}^2 \le (s - A_*)^2 + \sum_{k=0}^{i-2} 4^k \le (s - A_*)^2 + \frac{A_*^2}{3} \le \frac{s^2}{3}, \tag{4.1}$$

$$\sum_{k=0}^{i-1} e_{i,k} \le 2^i - 1 - i. \tag{4.2}$$

The last inequality in (4.1) is equivalent to $(s - A_*)(s - 2A_*) \le 0$. Define

$$B_i = i\eta_i + 4\eta_i \sum_{k=0}^{i-1} e_{i,k} + 4(2^i - 1).$$

Because $4s^2 \le n_i^2$, equation (3.1) implies

$$L(n) \ge \sum_{i=1}^{r} (i - 1/3) 4^{r-i} n_i^2 - \sum_{i=1}^{r} 4^{r-i} B_i. \tag{4.3}$$

The prefix identity $0 \le n - 2^{r-i} n_i < 2^{r-i}$ gives

$$\begin{aligned} \sum_{i=1}^{r} (i - 1/3) 4^{r-i} n_i^2 &\ge n^2 \left(\frac{r^2}{2} + \frac{r}{6} \right) - 2n2^r \sum_{i=1}^{\infty} (i - 1/3) 2^{-i} \\ &= n^2 \left(\frac{r^2}{2} + \frac{r}{6} \right) - \frac{10}{3} n 2^r. \end{aligned}$$

By (4.2) and $0 \le \eta_i \le 1$,

$$\begin{aligned} \sum_{i=1}^{r} 4^{r-i} B_i &\le 4^r \sum_{i=1}^{\infty} 4^{-i} \{ i + 4(2^i - 1 - i) + 4(2^i - 1) \} \\ &= 4 \cdot 4^r. \end{aligned}$$

Substitution in (4.3), followed by $2^r \le n$, proves (3.3). If $n = 2^r$, all remainders and all η_i vanish. Summing $K_i = i4^i - 4(2^i - 1)$ gives (1.4).

For any integer $D \ge 3$ take $n = \lfloor (D + 1)/2 \rfloor$. Then $2n - 1 \le D$, $n = D/2 + O(1)$, and $r = (\ln D)/(\ln 2) + O(1)$. Equation (3.3) yields (1.3). In particular the claimed coefficient is an all-degree lower bound, rather than only a subsequence coefficient. This completes the proof of Theorem 1.1.

5 Comparison with the preceding constructions

5.1 Two-source renewal and mechanical histories

Both the renewal theorem of [8, Proposition 4.1] and the present construction use fourfold copying with two axis sources. The earlier theorem starts from an arbitrary polynomial field

carrying a selected finite hyperbolic configuration. It transplants that configuration into the interior of a square and rebuilds the traces on two adjacent edges at a cost of two degrees. A signed tangency condition makes all simple source roots available for center creation. The classical fourfold transformation and two successive common perturbations then give the update stated in the introduction. The normalized number of new cycles per stage is of constant order, producing the single logarithm.

Proposition 3.3 instead constructs the separable Hamiltonians and perturbing forms together. Its source at stage i retains i auxiliary birth levels, so the normalized new-zero count is of order i. Summation over the planar stages supplies the second logarithm. The future-degree allocation pays for this larger source. Exact cancellation uses a reflection fixing each new oval and reversing its orientation; exchanging two different ovals would only make their periods opposite. Small even additions preserve this cancellation while filling the intervening degrees.

Thus the two results address different input classes as well as different quantitative bounds. The earlier renewal preserves a given arbitrary hyperbolic configuration. The present theorem improves the asymptotic count along an explicitly constructed near-Hamiltonian sequence; it does not assert the same improved update for arbitrary input fields. The proof constructs the auxiliary history and global form directly, without the transplantation or signed edge traces used in the earlier theorem.

5.2 The Qin–Sun construction and all-degree comparison

Qin–Sun's two logarithms arise from two 3-adic filtrations, one in each coordinate, in an anisotropic Chebyshev center grid, with a simultaneous choice of perturbation coefficients. Here they arise from the auxiliary mechanical history and the stages of planar renewal. The square-logarithmic order is therefore a prior result of Qin–Sun; the contribution here is the mechanical construction and the improved coefficient. To compare the coefficients over the same range of degrees, we first transfer their subsequence bound to all sufficiently large degrees.

The required replication operation is due to Eshkobilov–Kadyrov–Mamayusupov [2, Theorem 1, Lemma 2(4) and Section 6.3], whose time-change argument also preserves hyperbolicity. We record the form used here.

Lemma 5.1. *If a real polynomial field of degree at most d has N selected hyperbolic limit cycles, then for every integer $m \geq 1$ there is a real polynomial field of degree at most $m(d+1)-1$ with at least $m^2 N$ hyperbolic limit cycles.*

Proof. An affine change of coordinates places the selected compact cycles inside $(-1,1)^2$. Write the resulting field as (P, Q) and let T_m be the first-kind Chebyshev polynomial. The polynomial field

$$\big(T_m'(y)P(T_m(x), T_m(y)),\ T_m'(x)Q(T_m(x), T_m(y))\big)$$

has the stated degree. Each of the m^2 open monotonicity rectangles maps diffeomorphically onto $(-1,1)^2$. Pushing the field forward gives the seed field multiplied by the nonvanishing scalar $T_m'(x)T_m'(y)$. Thus each selected cycle has one copy in every rectangle. A positive time change preserves its return map, and a negative time change replaces that map by its inverse. The multiplier is consequently either the original multiplier or its reciprocal; hyperbolicity persists in both cases. □

Combining the lemma with (1.2) gives the explicit bound

$$H_{\mathrm{hyp}}(D) \geq B_{\mathrm{QS}}(D) := \max_{\substack{r\geq 2\\ 4\cdot 3^r-5\leq D}} \left\lfloor \frac{D+1}{4\cdot 3^r-4} \right\rfloor^2 [4r^2 3^{2r-2} - (3^r-1)^2] \qquad (D \geq 31). \tag{5.1}$$

The case $m = 1$ in Lemma 5.1 recovers each unreplicated seed; the factor in (5.1) chooses the largest admissible m for each seed degree.

Proposition 5.2. *For all sufficiently large integer D,*

$$H_{\mathrm{hyp}}(D) \geq \frac{D^2}{36(\ln 3)^2}\big[(\ln D)^2 - 2\ln D \ln\ln D\big] - O(D^2 \ln D). \tag{5.2}$$

Proof. Put $S = D + 1$ and choose r maximal with $4 \cdot 3^r \leq S/\ln S$. Then $q = 3^r = \Theta(S/\ln S)$ and

$$r = \frac{\ln D - \ln\ln D + O(1)}{\ln 3}.$$

Set $m = \lfloor S/(4q-4) \rfloor$. This meets the degree bound and $mq = D/4 + O(D/\ln D)$. Inserting these estimates in $m^2[(4/9)q^2r^2 - (q-1)^2]$ gives (5.2); the remaining $(\ln\ln D)^2$ term is $O(\ln D)$. □

For clarity, the relevant coefficients are displayed in Table 1. Qin–Sun's direct all-degree formula [9, Corollary 1.2 and Sections 6.2, 6.4] gives, for every $D \geq 31$,

$$H_{\mathrm{hyp}}(D) \geq \frac{(D+5)^2}{324}\left(\frac{\ln((D+5)/12)}{\ln 3}\right)^2 - \frac{(D+1)^2}{16}.$$

Its smaller coefficient comes from filling gaps between consecutive ternary seed degrees by monotonicity. Proposition 5.2 is the consequence of their seeds and replication proved here, and is not a restatement of that corollary.

Result	Degree range	Coefficient c	Error factor
Qin–Sun, Theorem 1.1	$4 \cdot 3^r - 5$, $r \geq 2$	$1/[36(\ln 3)^2]$	$\ln D$
Qin–Sun, Corollary 1.2	Every $D \geq 31$	$1/[324(\ln 3)^2]$	$\ln D$
Qin–Sun + replication, Proposition 5.2	Every sufficiently large D	$1/[36(\ln 3)^2]$	$\ln D \ln\ln D$
Mechanical renewal, Theorem 1.1	Every sufficiently large D	$1/[8(\ln 2)^2]$	$\ln D$

Table 1: Bounds of the form $H_{\mathrm{hyp}}(D) \geq cD^2(\ln D)^2$ with remainder $O(D^2)$ times the displayed error factor. All degrees are total degrees of one real planar polynomial vector field; all counted cycles are hyperbolic.

6 An iterated renewal conjecture

The preceding constructions suggest the following sufficient scheme for higher logarithmic powers. Its unresolved step is the simultaneous dynamical realization of new and inherited cycles at controlled degree.

Conjecture 6.1 (Iterated renewal)**.** *For each fixed integer $k \geq 3$, there are integers $b, A \geq 2$ and a constant $c > 0$, all depending only on k, with the following property. For every integer $r \geq 1$ there is a finite sequence of real planar polynomial fields $X_0, \ldots, X_r$. For $0 \leq j \leq r$, $\deg X_j \leq Ab^j - 1$ and X_j carries a selected finite family of N_j hyperbolic limit cycles. At each step $1 \leq j \leq r$, this family contains the hyperbolic continuations of all b^2 pullback copies of each selected cycle of X_{j-1} and at least $cb^{2j}j^{k-1}$ additional cycles. The pullback is separable of degree b, with b real diffeomorphic inverse branches in each coordinate over the preceding configuration. All continued and additional cycles are pairwise distinct in X_j.*

The fields and auxiliary parameters may depend on the prescribed finite horizon r, as in Proposition 3.3. The input fields are to be constructed as part of the sequence; the conjecture does not require an update accepting every polynomial input field. A hyperbolic continuation means that each selected pullback cycle persists hyperbolically along a deformation from the pullback field to X_j through polynomial fields of degree at most $Ab^j - 1$; its multiplier may vary.

The claimed construction would give

$$N_j \geq b^2 N_{j-1} + cb^{2j}j^{k-1}, \qquad \frac{N_r}{b^{2r}} \geq c\sum_{j=1}^{r} j^{k-1} \geq \frac{c}{k}r^k.$$

For large D, taking $r = \lfloor \log_b((D+1)/A) \rfloor$ gives $Ab^r - 1 \leq D$, $b^r > (D+1)/(Ab)$ and $r \sim \ln D / \ln b$. Thus the conjecture would imply

$$H_{\mathrm{hyp}}(D) \geq c_k D^2 (\ln D)^k$$

for some $c_k > 0$ and all sufficiently large D. Degreewise existence of a field with a lower logarithmic power does not itself provide the renewal asserted in Conjecture 6.1: the new cycles must be created together with the retained copies at every stage without an additional degree cost that destroys the estimate. The constructions for one and two logarithms motivate this question; no such iterated realization for $k \geq 3$ is proved here.

Reproducibility and use of AI

The accompanying program `repro/verify.py` checks the finite counts, degree inequalities and replication envelope by integer arithmetic. These checks supplement the analytic proofs above.

OpenAI's ChatGPT and Codex were used to explore proof strategies, check calculations, develop verification scripts, and revise the text. The author is responsible for all mathematical claims and the final manuscript.

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